

MAR 6930 · Marketing Data Visualization

Data exploration

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Goals

1. Summarize data and make comparisons: pivot tables and pivot charts.
2. Understand more about a variable from its distribution: histogram, box plot, percentile chart, Pareto chart and Lorenz curve.
3. Common traps in visualization

Data summary and comparison

Pivot table and pivot chart

Pivot tables turn raw rows into a summary table

Definition: A pivot table rearranges raw data into rows, columns and values to summarize how a variable differs across groups.

Raw data: a retailer's sales over 20 weeks, one row per sales record (95,100 rows)

Date	Store	Channel	Category	Revenue
2025-01-01	1	In-Store	Beauty	\$13.21
2025-01-01	1	In-Store	Beauty	\$17.10
2025-01-01	1	Online	Household	\$9.71
2025-01-01	1	Online	Household	\$10.17
2025-01-01	1	Online	Snacks	\$3.19
2025-01-01	1	In-Store	Health	\$6.70

Pivot table: revenue by week and channel (20 weeks × 2 channels)

Week	In-Store	Online
1	\$33,026	\$15,450
2	\$33,705	\$14,607
3	\$33,874	\$14,717
4	\$34,761	\$15,338
5	\$37,236	\$17,505

Takeaway: From one raw table you can build many pivot tables: by grouping (week, store, category), by statistic (sum, average, proportion), by filter (only sales records over \$30).

Three kinds of pivot table

One-way pivot table

Summarizes a variable by one grouping variable.

Region	Revenue
Northeast	\$456k
South	\$247k
West	\$212k
Midwest	\$184k

Grouping variable: region (rows)

Two-way pivot table

Summarizes a variable by two grouping variables.

Region	In-Store	Online	Total
Northeast	\$303k	\$153k	\$456k
South	\$176k	\$71k	\$247k
West	\$142k	\$70k	\$212k
Midwest	\$133k	\$52k	\$184k
Total	\$755k	\$345k	\$1,100k

Grouping variables: region (rows) and channel (columns)

Totals: a two-way or multi-way table can add a total at the end of each row and column; the Total column is the one-way table again.

Multi-way pivot table

Uses three or more grouping variables, usually with multiple two-way pivot tables for different categories.

Beauty	In-Store	Online
Northeast	\$112.8k	\$57.4k
South	\$65.2k	\$27.6k
West	\$55.4k	\$26.6k
Midwest	\$50.5k	\$20.5k
Snacks	In-Store	Online
Northeast	\$15.0k	\$7.9k
South	\$9.5k	\$3.4k
West	\$7.5k	\$3.3k
Midwest	\$6.3k	\$2.5k

Grouping variables: region (rows), channel (columns) and category (one table each)

A pivot table can summarize any statistic you need

The cell value is whatever answers the question: a sum, an average, a proportion, or a combination of variables or their statistics.

Example: a one-way table (by region) with a different statistic in each column

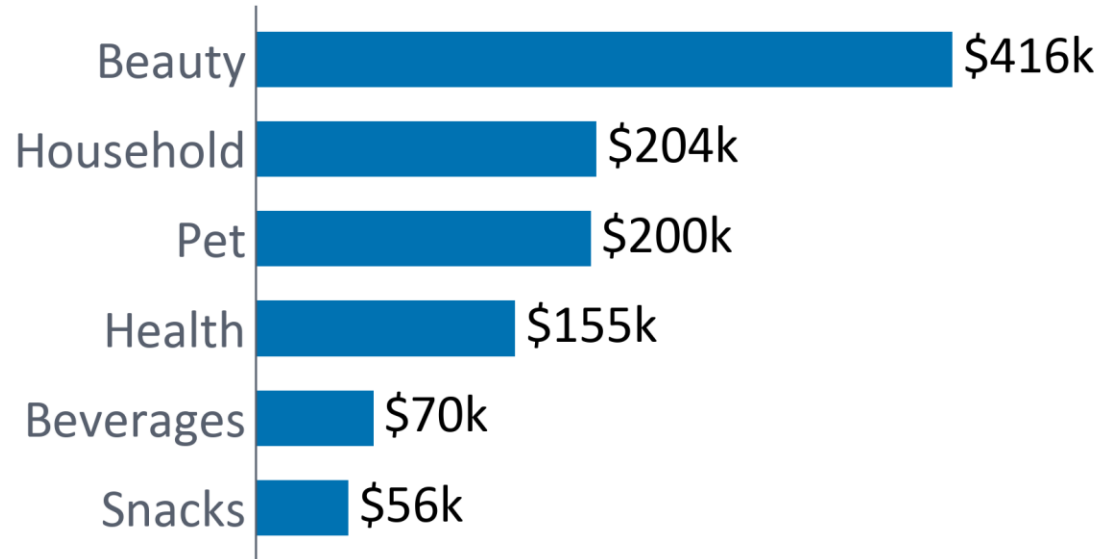
Region	Revenue	Revenue per sales record	Margin = revenue – cost	Online share	Online share, only sales records with margin ≥ \$10	Best week revenue (max) ÷ worst week (min)
	(sum)	(average)	(relationship between two variables)	(proportion)	(data filtering conditions)	(max ÷ min: relationship between two stats of a variable)
Northeast	\$456k	\$11.53	\$210k	33.5%	34.4%	1.28
South	\$247k	\$11.57	\$114k	28.7%	29.8%	1.33
West	\$212k	\$11.57	\$98k	33.0%	33.5%	1.50
Midwest	\$184k	\$11.67	\$85k	32.8%	33.8%	1.25

Takeaway: Create the pivot tables that help answer your business question, with proper choices of categories, statistics, and data filtering conditions.

Visualize pivot tables with pivot charts

From a one-way table: a bar chart compares the categories

**An amount:
total revenue by category, Jan–May**

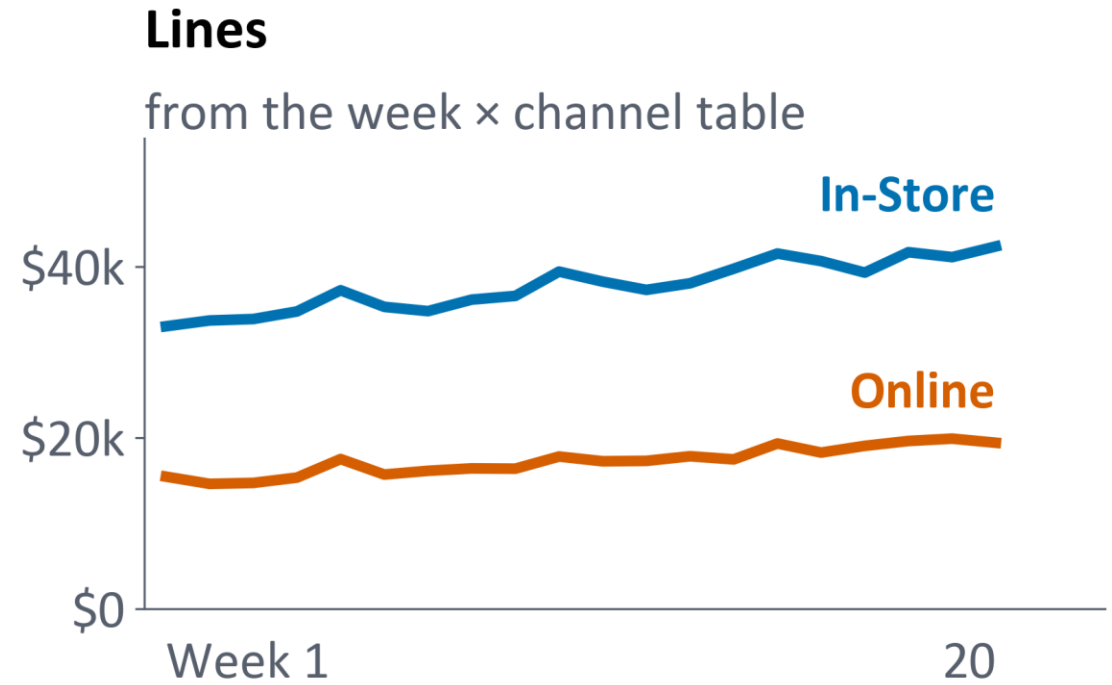
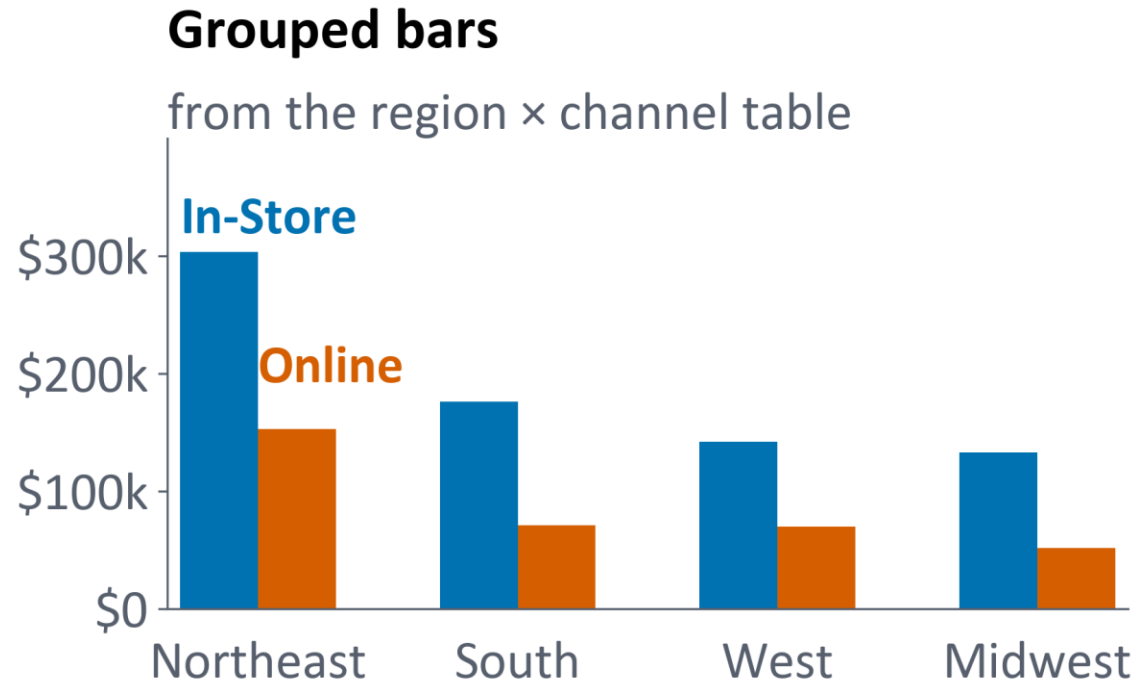


**A percentage:
online share of revenue by region**



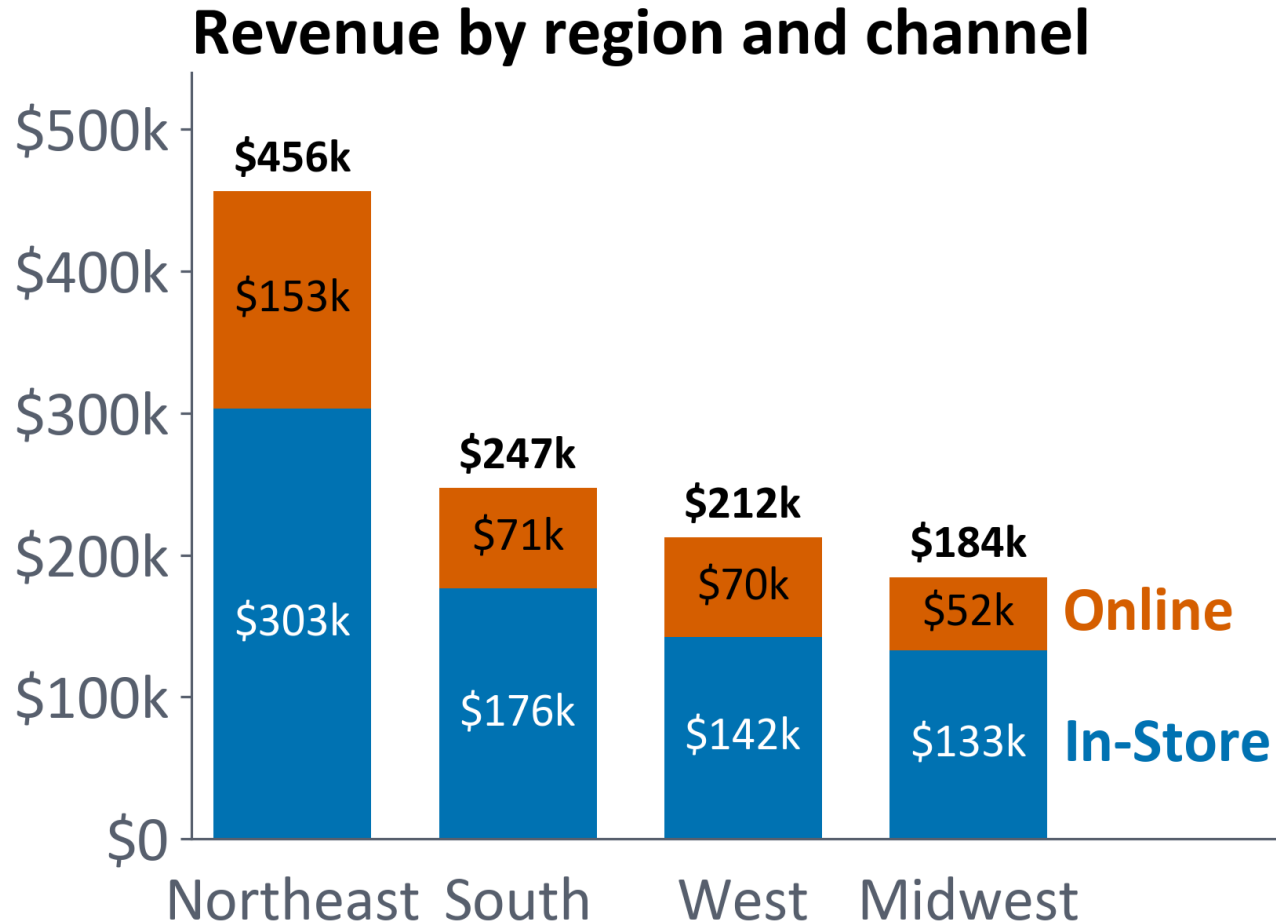
Ideally, rank the categories by size when they have no order of their own: the ranking then reads at a glance.

From a two-way table: grouped bars or lines



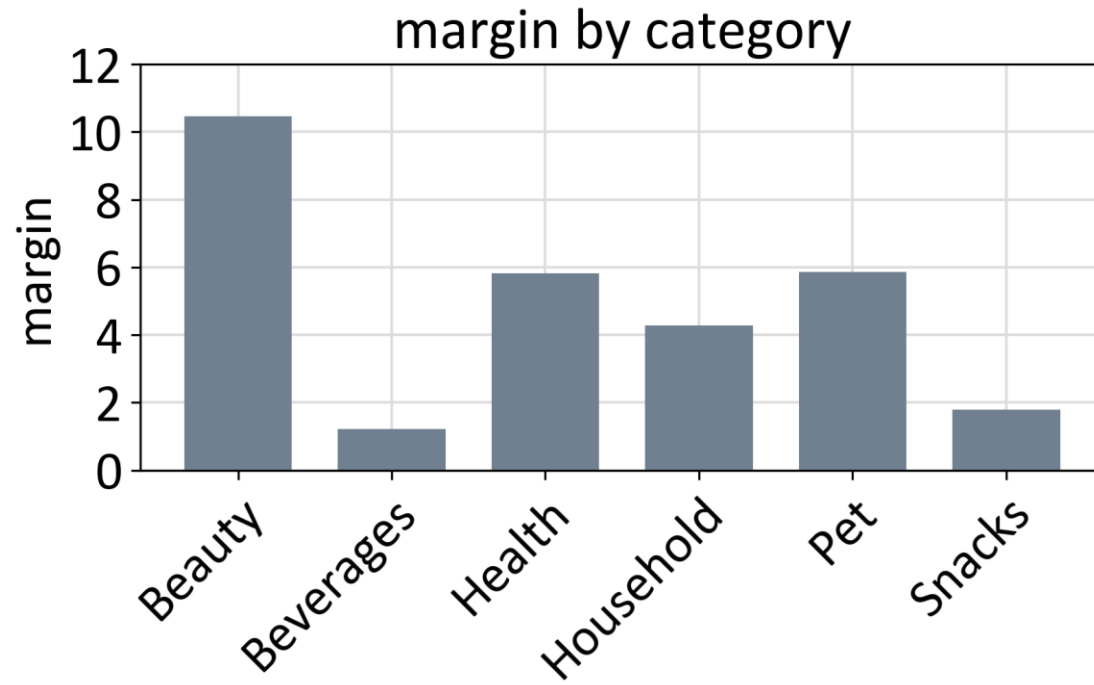
- Grouped bars compare the groups inside each category.
- Lines show the trend when one grouping variable is time.
- Caution: too many groups create clutter. Picture how messy the chart would be with 10 bars per group; if you must show them all, highlight only the bars that carry your message.

Stacked bars: the totals, and how each one splits

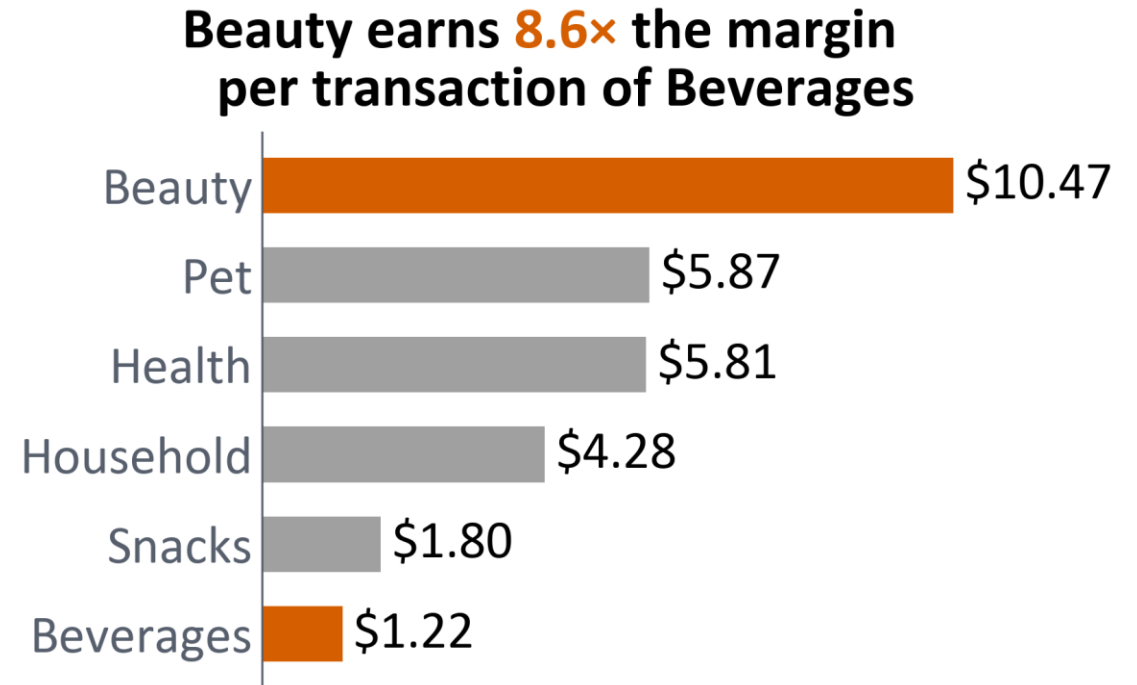


- Use stacked bars when comparing the total matters, but differences between groups (online) become harder to see.
- Use grouped bars to compare the parts.

From exploration to presentation: recreate the chart to highlight the main message



Exploring: minimal formatting and many quick charts



Presenting: once you know the message, rebuild the chart: rank the categories by the metric, highlight the comparison, and write the message.

Build these summaries in Python

Notebook: lecture3-part1-summary-comparison.ipynb, with L3-retail.csv

- The raw rows and the pivot table (revenue by week and channel)
- The region table with one statistic per column
- Bars of category totals and of the online share by region
- Grouped bars, lines and stacked bars from the two-way tables
- A three-way table: weekly revenue by region, one chart per channel
- The exploration and presentation versions of margin per sales record by category

Exercise: which stores earn the most margin, and how does one store's margin move?

Notebook: [lecture3-part1-summary-comparison.ipynb](#), exercise at the end of the notebook

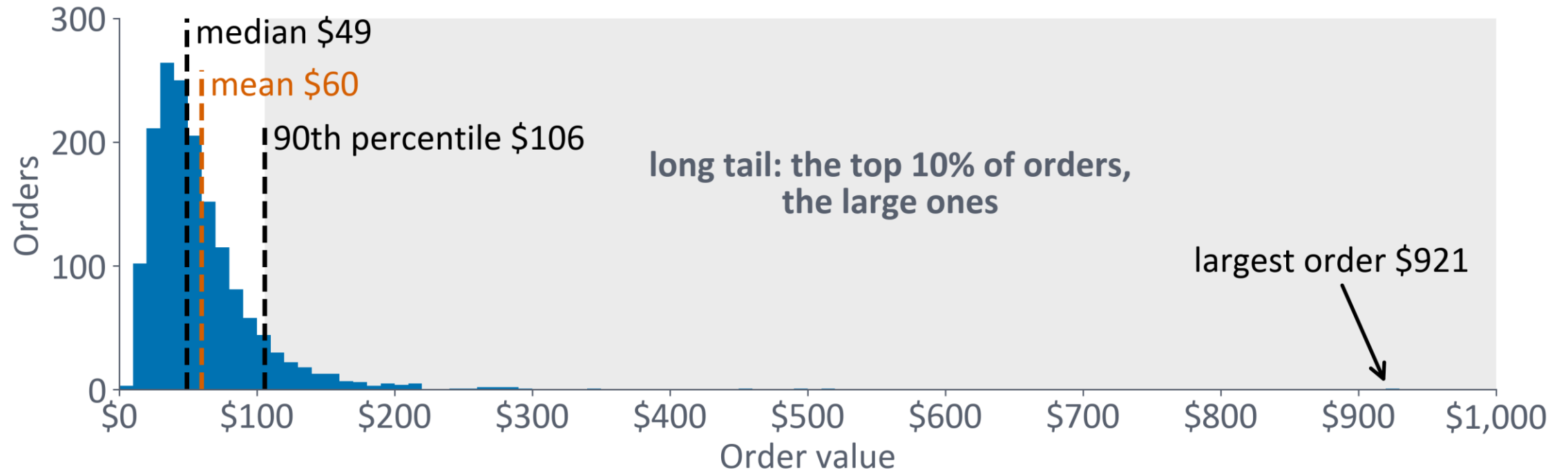
Zooming in on a variable's distribution

Is the average order a typical order?

Average order value: **\$60**

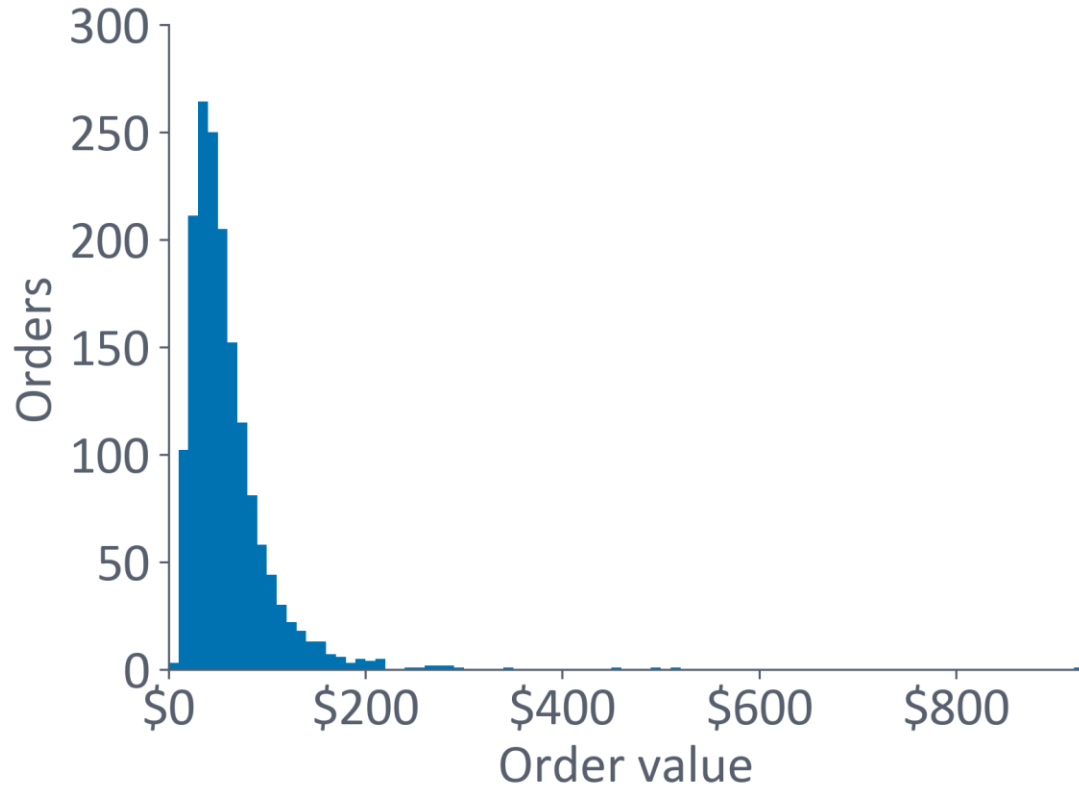


The data distribution tells more than just the average



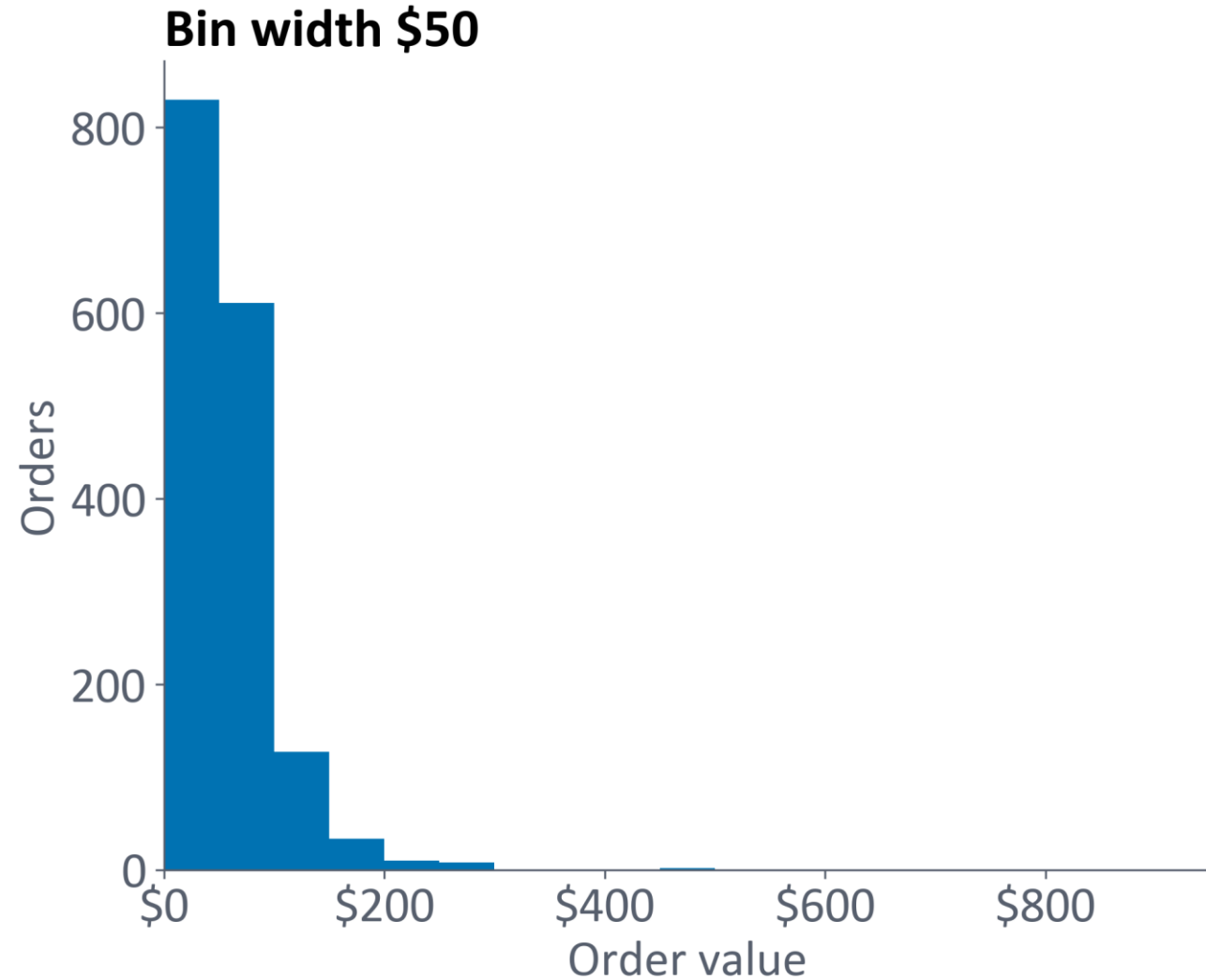
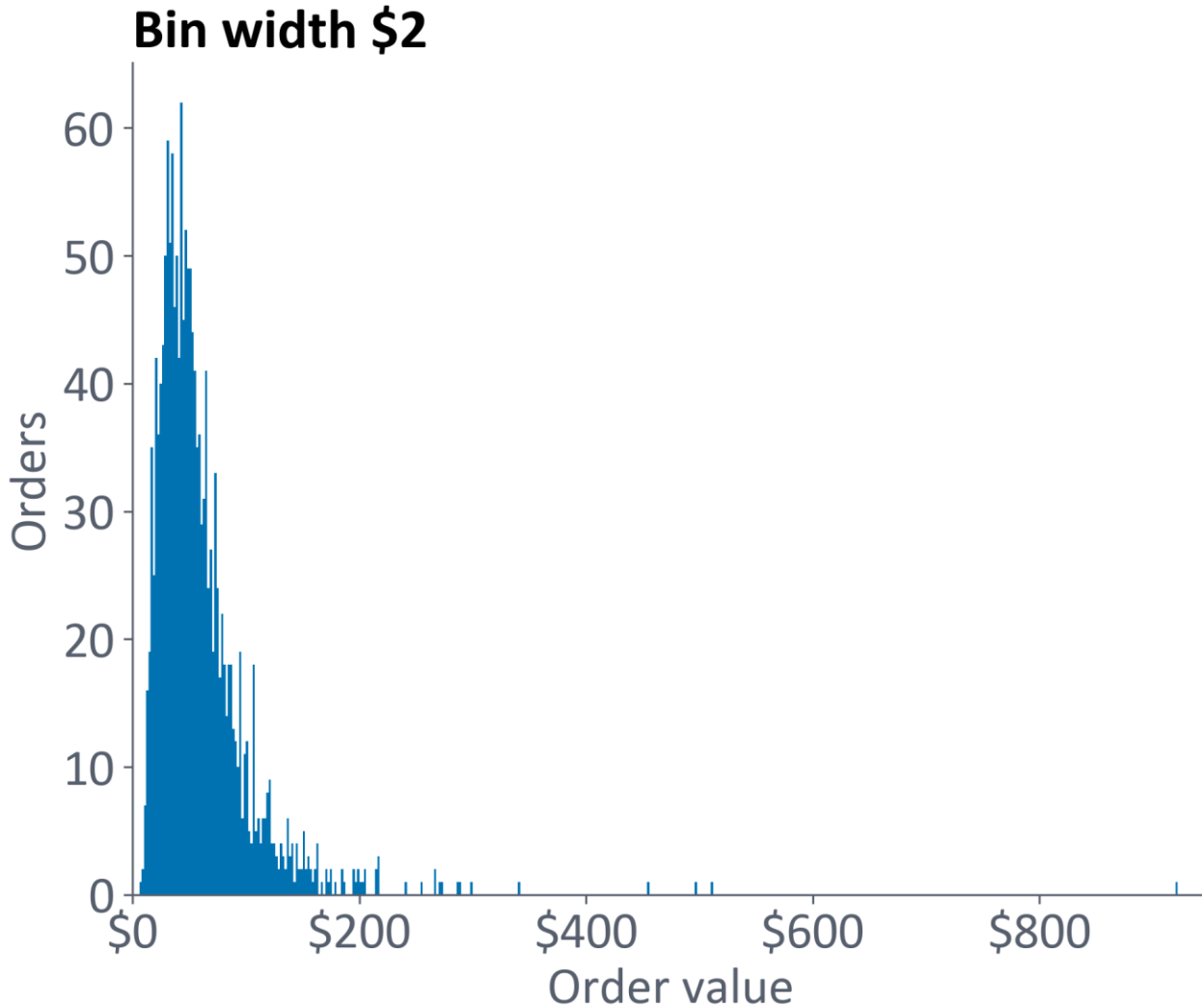
- Mean \$60, median \$49: half of the orders are below \$49, and the tail pulls the mean up.
- The top 10% of orders are above \$106; the largest is \$921.

Histogram: the shape of the distribution



- **Definition:** cut the value range into equal bins and count the orders in each; bar height = count.
- One peak around \$35–50, then a long right tail: a few orders run to \$921.
- Skewed to the right, so the mean (\$60) sits above the median (\$49).

Same orders, two bin widths: which shape is real?



Which summary number fits the decision?

Question 1: what is a typical case like?

Mean

Best indicator for totals (revenue = mean order × number of orders).
Sensitive to extreme outliers: e.g., the average income in the town where Elon Musk lives.

Median

The typical case.
Not sensitive to outliers.

Discussion: if the mean and the median differ a lot, beware of extreme values or a long tail.

Question 2: how do the extreme cases in the tail behave?

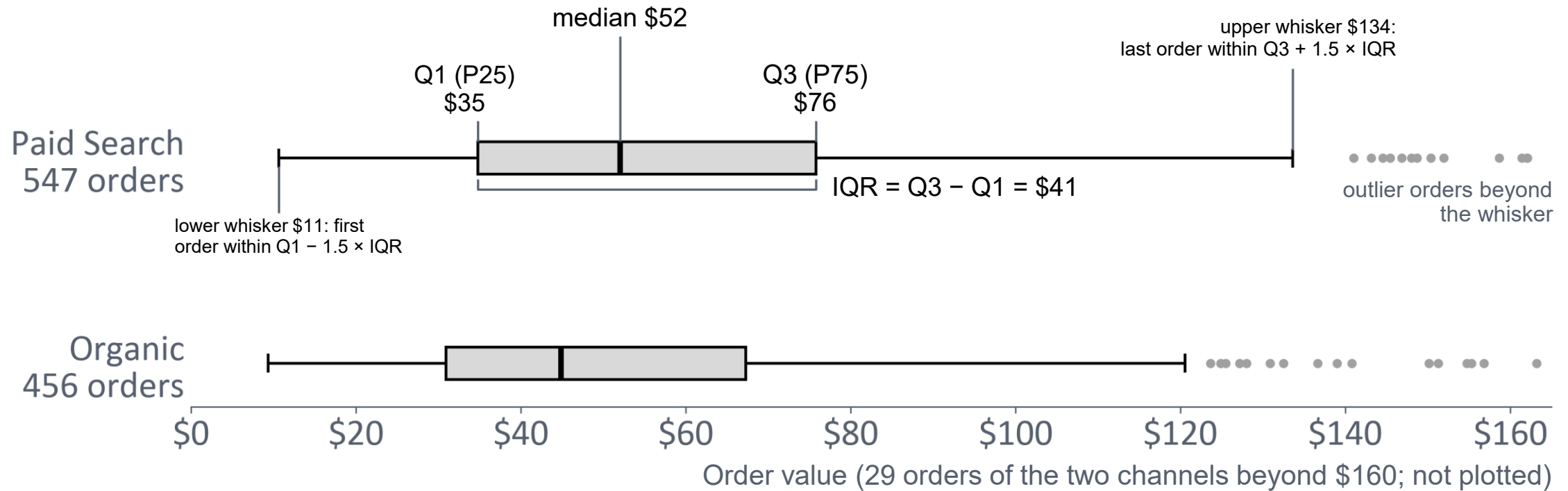
Percentiles (P90, P99)

Guarantees, tiers and risk limits.
e.g., a support team promises “answered within 4 minutes for 90% of calls” (P90, not the average).

Tail share (what the top X% bring)

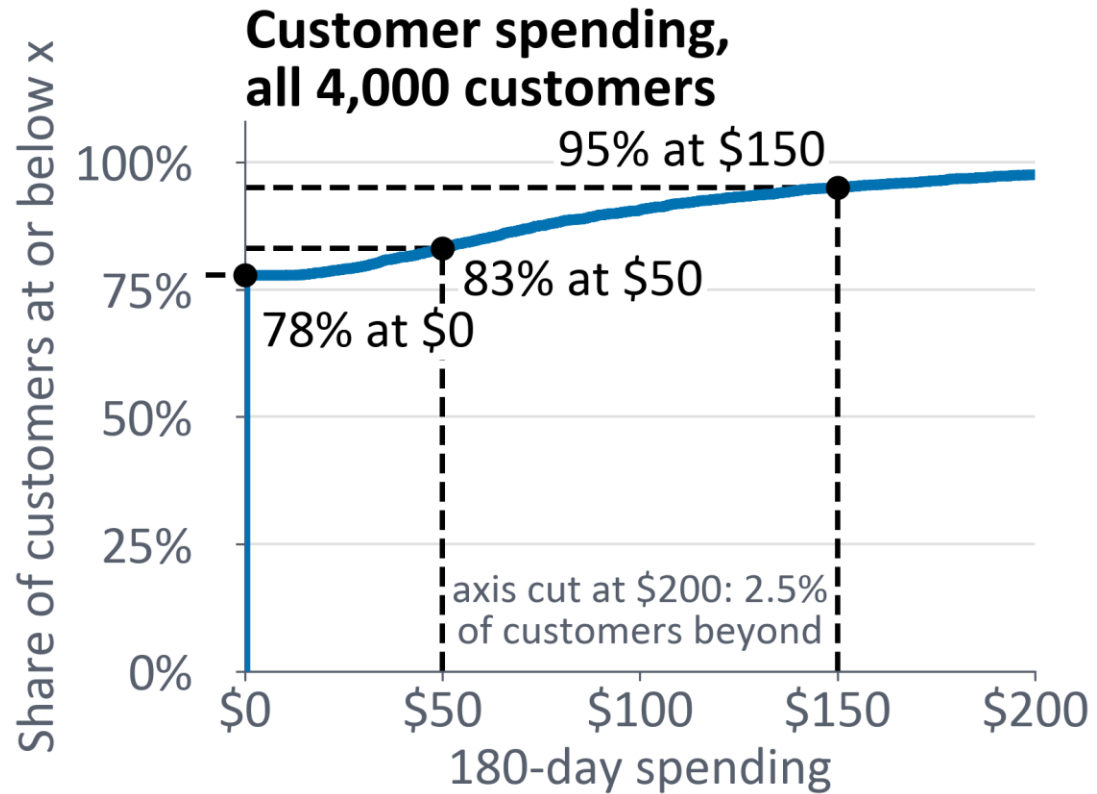
Concentration and priorities.
e.g., “80% of profit comes from the top 20% of customers.”

Box plot: the middle half, the median and the tails



- **Definitions:** the box runs from Q1 (P25) to Q3 (P75); its length is the **interquartile range**, $IQR = Q3 - Q1$, and the dividing line is the median.
- **Whisker definition:** software differs in whisker conventions: some indicate the minimum and the maximum; others (including common Python tools) indicate the last value within $1.5 \times IQR$ of the box.
- Specify your definitions of whiskers in your box plots.

Percentile chart: what share lies below a value?



- **Definition:** the percentile chart (also called the ECDF, the empirical cumulative distribution function) shows, for each value x , the share of cases at or below x .
- At \$0 the curve already stands at 78%: 3,111 of the 4,000 customers bought nothing in 180 days, so the median customer spent \$0.
- At \$50 it reaches 83%; at \$150, 95%: one customer in twenty spent more than \$150.

From a target share to a cutoff value

Pick the share first, then read the value off the curve: a percentile is the percentile chart read backwards.

VIP tier = the top 10% of spenders → the cutoff is P90 of customer spending.

Risk review = the top 1% of orders → P99 of order value, \$217: review every order above it.

Capacity = “95% of orders are below \$134” → plan packing and stock for P95, not for the average.

ABC analysis: a few items carry most of the value

Definition: ABC analysis ranks items by value (e.g., profit generated) and splits them into three classes: A, the few items that carry most of the value; B, the middle group; C, the many small items.

- A items get tight control, accurate records and frequent counts; C items get the simplest rules.

Typical shares in inventory management

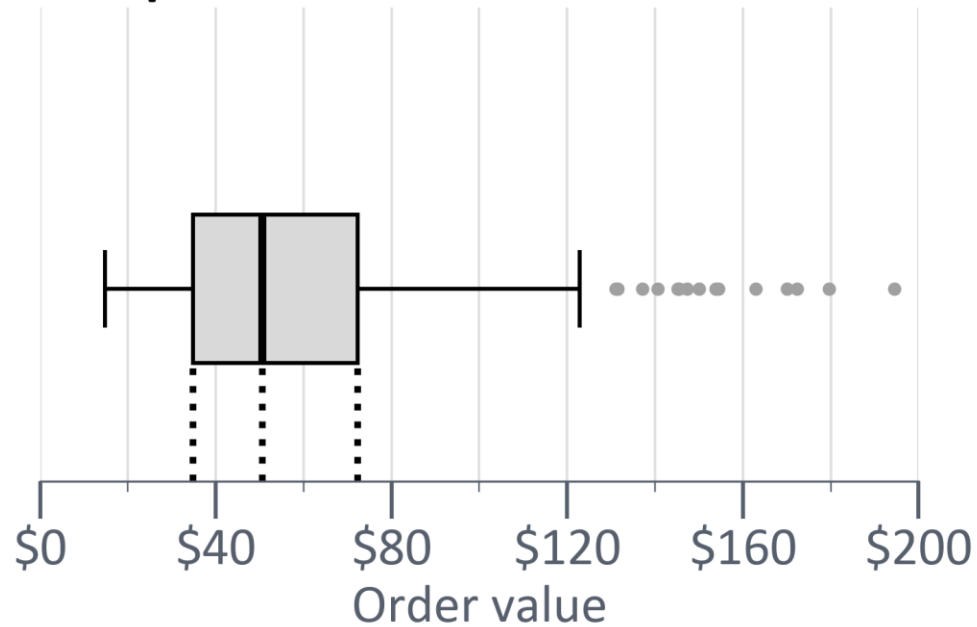
Class	Share of items	Share of annual value
A	top 20%	70%
B	next 30%	25%
C	bottom 50%	5%

Cutoffs from the percentile chart of item value: decide each class's share of items, then read the value at that share: the A cutoff at the 80th percentile, the B cutoff at the 50th.

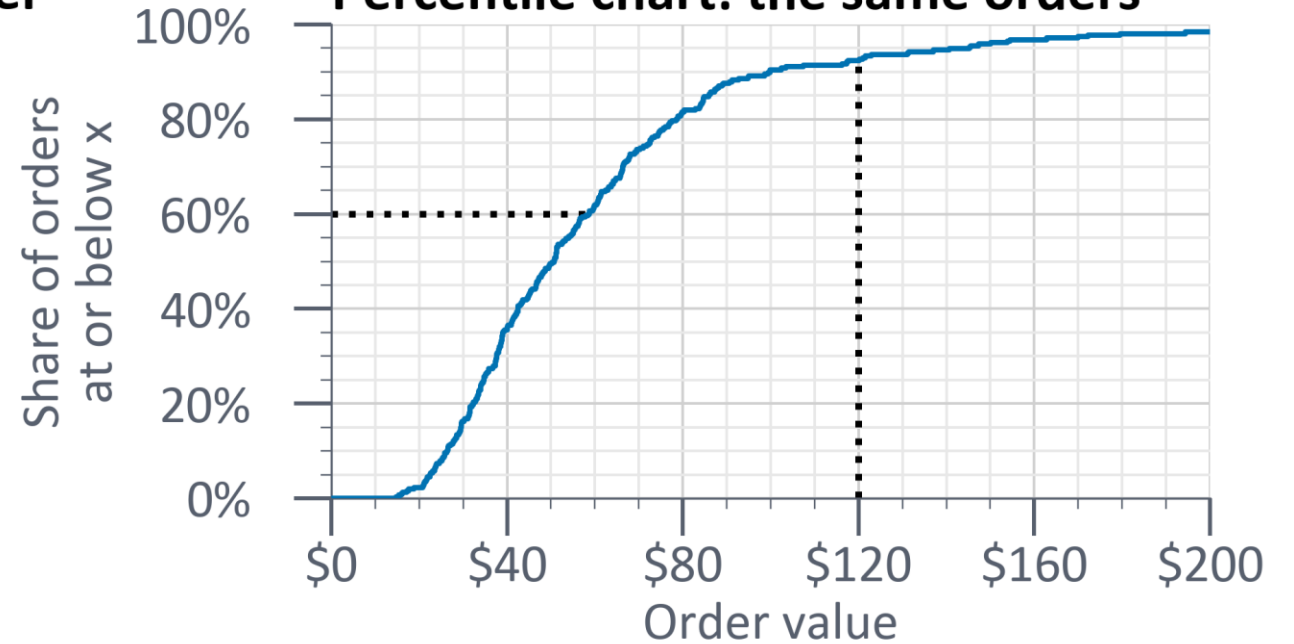
The same concept applies in many other settings, e.g., segmenting customers into three groups by the value they generate.

Exercise: read the box plot and the percentile chart

Box plot: orders from the Email channel



Percentile chart: the same orders



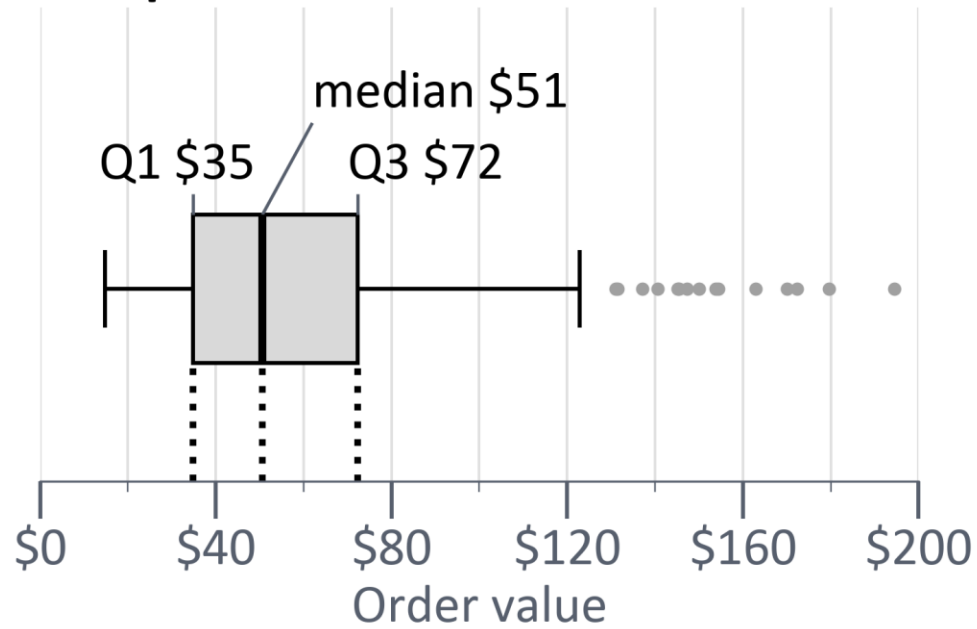
Box plot: what is the median order value, and between which two values does the middle half of the orders fall?

Percentile chart: what share of orders are at or below \$120, and below which value do 60% of orders fall?

Deliverable: four numbers.

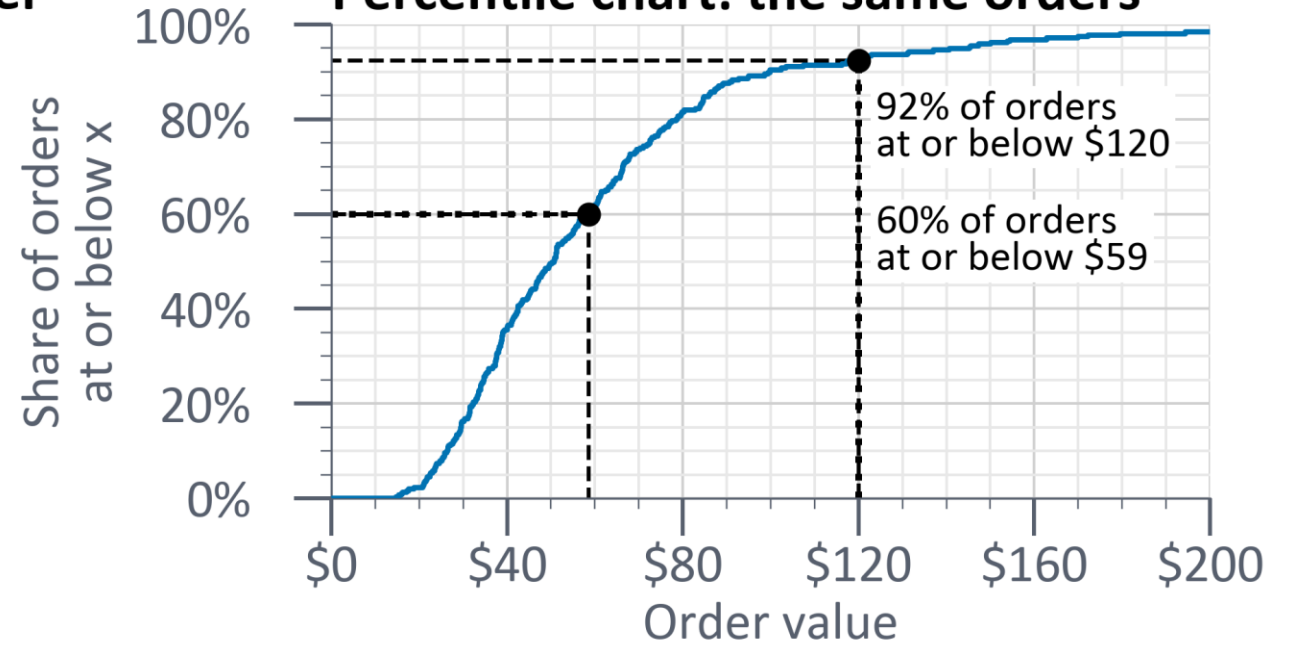
Reading the two charts: the answers

Box plot: orders from the Email channel



Median \$51; the middle half of orders runs from \$35 to \$72 (the box).

Percentile chart: the same orders



92% of orders are at or below \$120; 60% of orders are at or below \$59.

The 80/20 rule: a few causes bring most of the outcome

Pareto 80/20 principle: roughly 80% of outcomes come from 20% of causes or inputs.

- **Software quality:** about 20% of the bugs caused 80% of the reported errors at Microsoft; 1% caused half.
- **Health care:** the 5% of people with the highest spending account for about half of US health spending.
- **Wealth:** the top 1% of US households hold about 30% of household wealth.

The Pareto principle is a general description of uneven distribution, rather than a universal law: the real distribution can be more even (e.g., 70/30) or more uneven (e.g., 90/10).

Uneven distribution in marketing: most revenue may come from a small fraction of customers; most brand exposure may come from a small set of ads.

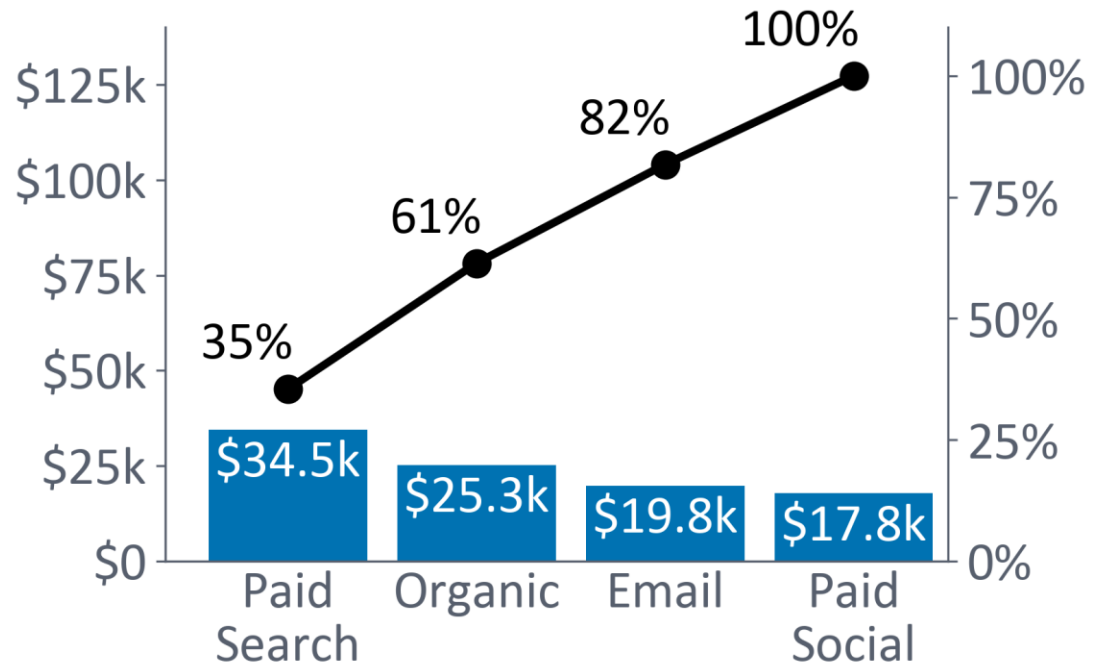
How do we measure such an uneven contribution?

Pareto chart, for categories: which channels, products or causes carry most of the total.

Lorenz curve, for a continuous variable: how spending spreads across customers.

Pareto chart: which categories carry most of the total?

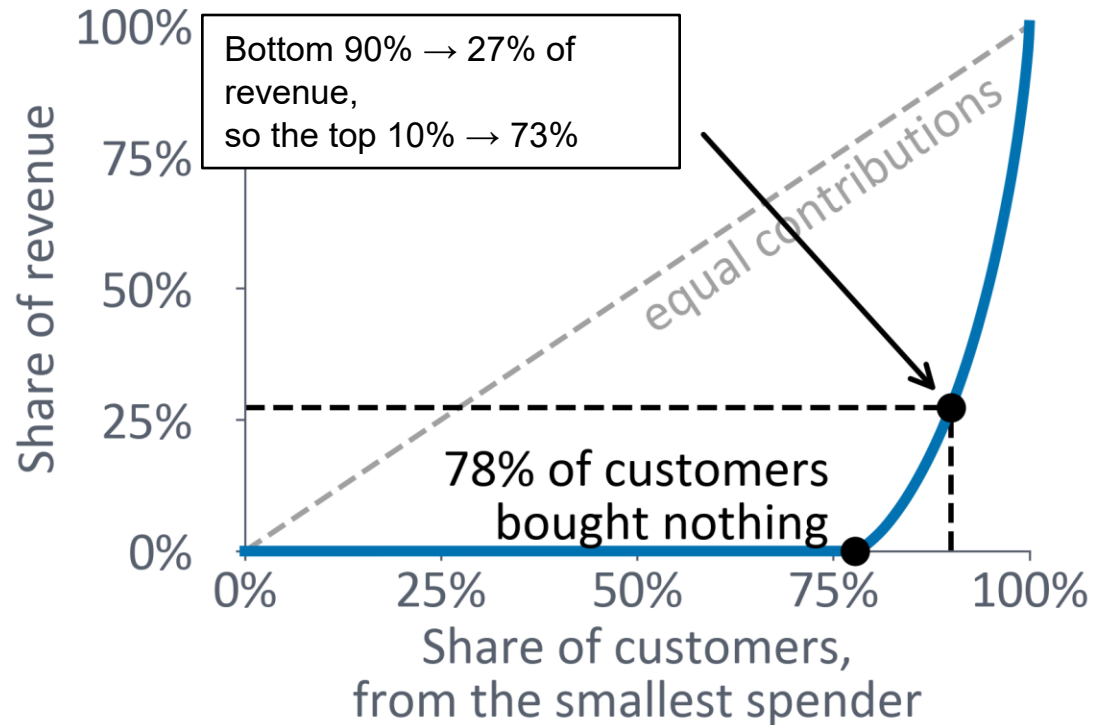
Revenue by acquisition channel,
with the cumulative share



- **Definition:** bars of a total by category, ranked from the largest, plus a line for the cumulative share: how many categories it takes to reach a given share of the total.
- Revenue by acquisition channel: Paid Search brings 35%; the top two channels 61%; the top three 82%.
- Classic use: the few causes behind most of the problems (complaints, defects, returns).

Lorenz curve: how is spending spread across customers?

Lorenz curve of customer spending



- **Definition:** for a continuous variable, sort the cases from the smallest; x is the share of cases counted so far, y their share of the total.
- The diagonal means equal contributions; the farther the curve bows below it, the more concentrated the total.
- The curve stays at 0 until 78% of customers: they bought nothing.
- The bottom 90% of customers bring 27% of revenue, so the top 10% bring 73%.

Build these distribution charts in Python

Notebook: `lecture3-part2-distribution.ipynb`, with `L3-transactions.csv` and `L3-customers.csv`

- Order values: the mean, the median and the percentiles
- Histograms at several bin widths
- Box plots of two channels, and two whisker rules
- The customer spending table, one row per customer
- The percentile chart, read forwards and backwards
- The Pareto chart of revenue by acquisition channel
- The Lorenz curve of customer spending

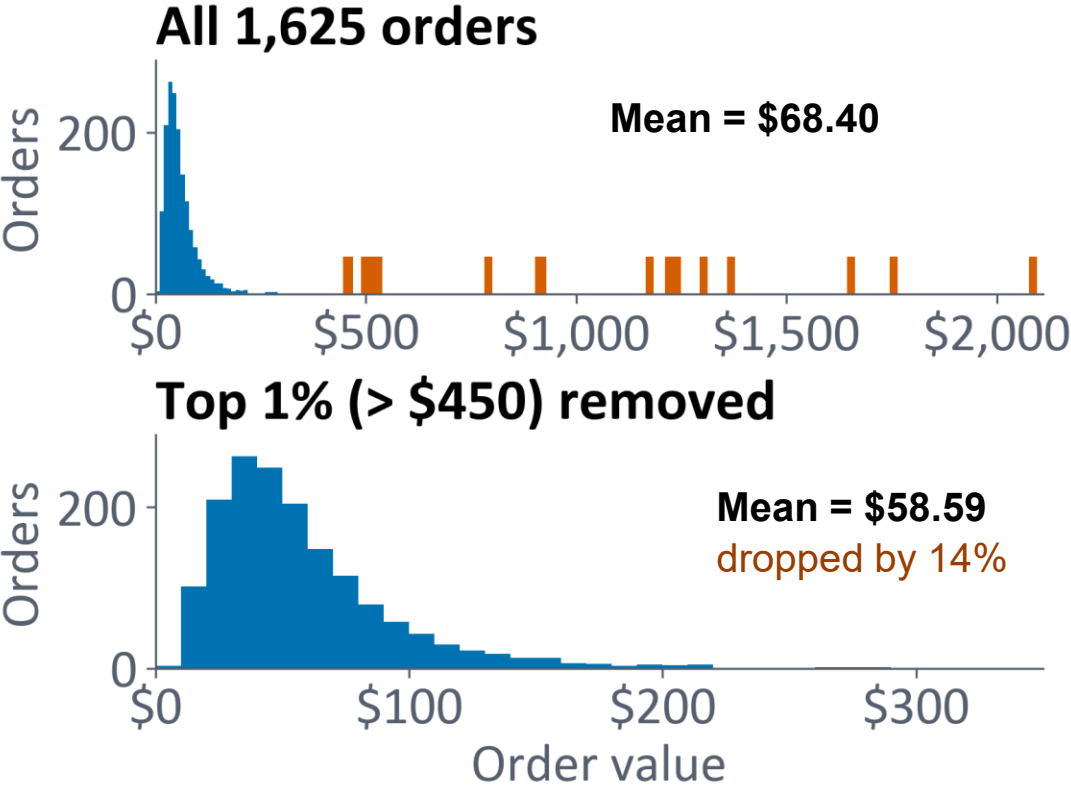
Exercise: who are the top 10% of customers, and how much margin do they bring?

1. From the orders, compute each customer's total margin; keep the customers with no purchases (4,000 customers).
2. Find the margin cutoff for the top 10% and mark it on the percentile chart.
3. Report the share of customers at or above the cutoff and their share of total margin; check it against the Lorenz curve at 90%.

Notebook: `lecture3-part2-distribution.ipynb`, exercise at the end of the notebook

Common issues of data treatment

An extreme outlier can hide the whole picture

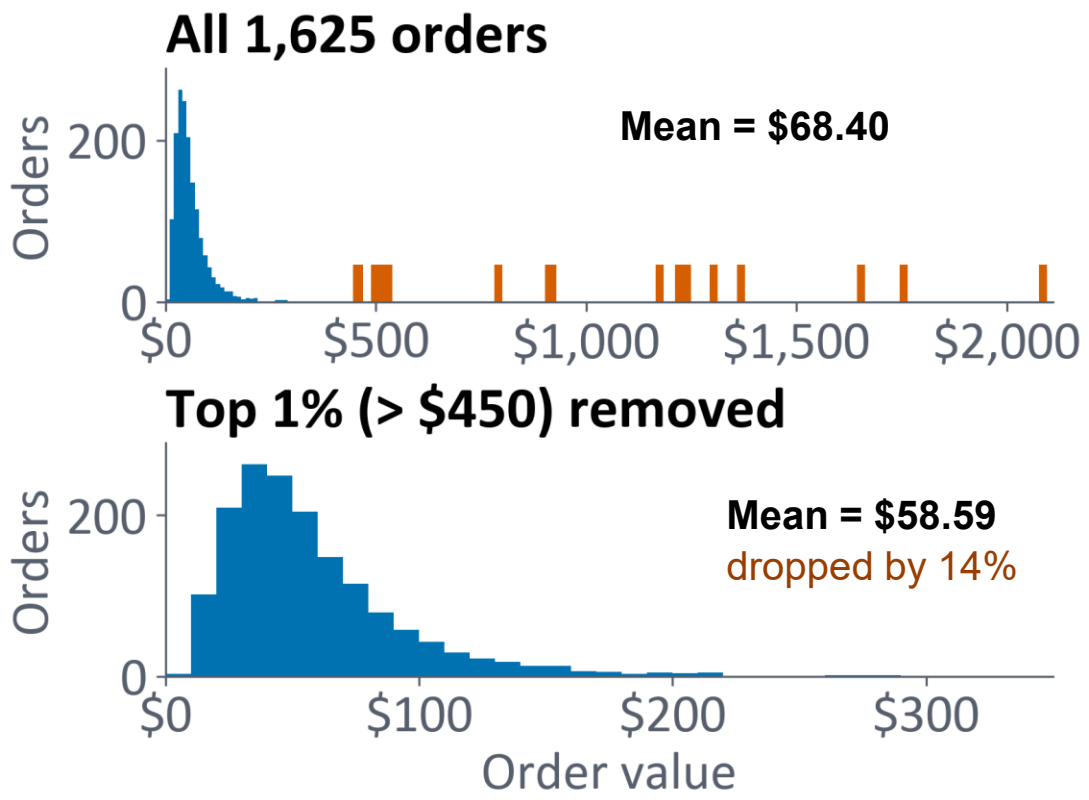


	All 1,625 orders	Top 1% removed	Change
Mean	\$68.40	\$58.59	-14%
Median	\$49.66	\$49.13	-1%

The mean is sensitive to outliers; the median is not.

With outliers, report both; each answers a different question:
“How much revenue does an average order bring?” → **mean**
“How large is a typical order?” → **median**

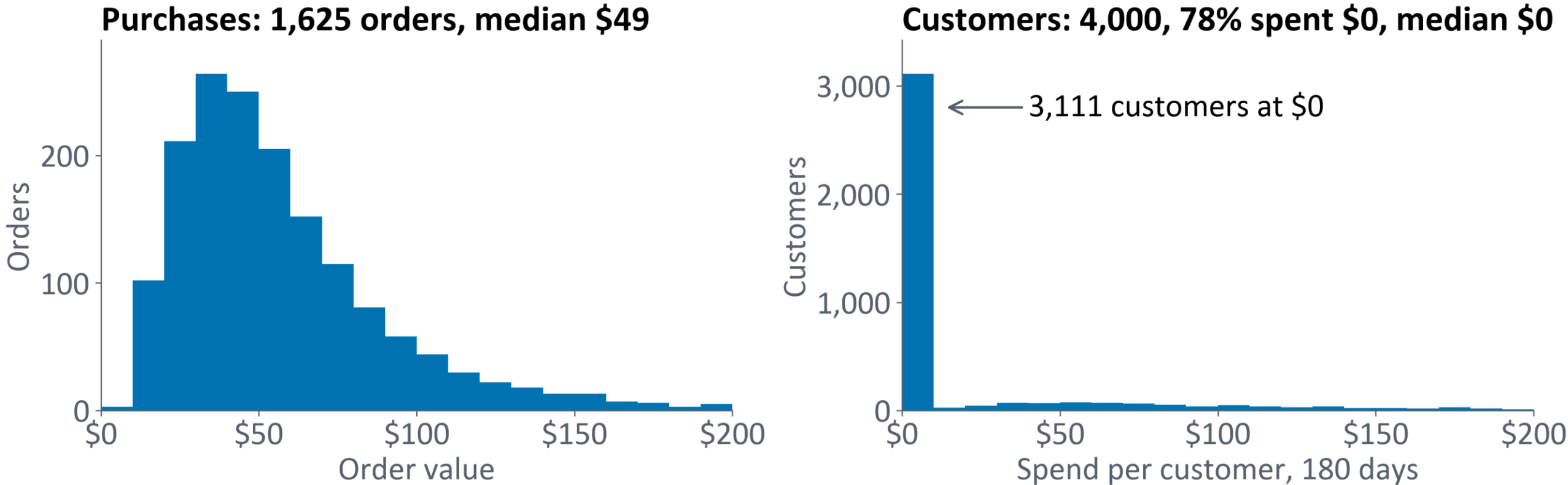
Outliers: when should we exclude them?



Remove / exclude when...	Keep when...
It is a data error: typo, impossible value, sensor failure	It is a real observation, even if rare
It comes from the wrong population: e.g., a B2B customer accidentally included in a consumer analysis	Extreme customers or events are part of the business you want to understand
A clearly different data-generating process produced it	The extreme value may contain useful information about the tail
Your research question explicitly excludes such cases	Removing it would mainly make the results “look nicer”

To show the shape of most orders, plot the bottom chart and note what it leaves out: "the top 1% of orders, which are above \$450."

Zero-heavy (sparse) data may obscure the visualization



Better graph: the left graph, with a note that only 22% of customers (889 of 4,000) bought anything; 78% spent nothing.

What is the margin rate of the whole store?

Category	Revenue	Margin	Margin rate
Beauty	\$400k	\$100k	25%
Snacks	\$100k	\$50k	50%
All	\$500k	\$150k	

X B is wrong: Beauty sells four times as much as Snacks, but averaging the rates weights the two equally.

Beauty: 2 of 8 bags = 25%



Snacks: 1 of 2 bags = 50%



Pooled: 3 of 10 bags = 30%



Each bag = \$50k of revenue; red = margin

Two ways to calculate it. Which is correct?

A: total margin ÷ total revenue = $(\$100k + \$50k) \div (\$400k + \$100k) = \$150k \div \$500k = 30\%$

B: average of the two margin rates = $(25\% + 50\%) \div 2 = 37.5\%$

When calculating the overall rate, do not average individual rates

Two groups, each with a rate = numerator ÷ denominator:

$$R1 = X1 / Y1, \quad R2 = X2 / Y2$$

$$R(\text{overall}) = (X1 + X2) / (Y1 + Y2)$$

$$\neq (X1 / Y1 + X2 / Y2) / 2$$

Here: $R(\text{overall}) = (\$100k + \$50k) / (\$400k + \$100k) = 30\%$, but $(25\% + 50\%) / 2 = 37.5\%$.

The two are equal only when $Y1 = Y2$ (or the rates are already equal).

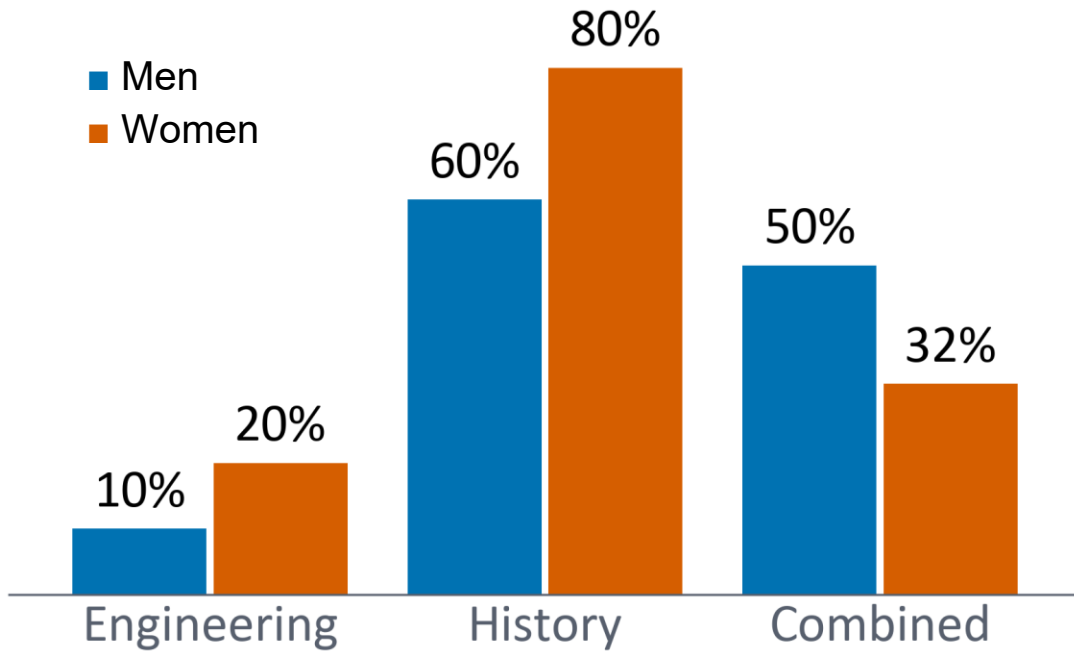
Category	Revenue	Margin	Margin rate
Beauty	\$400k	\$100k	25%
Snacks	\$100k	\$50k	50%
All	\$500k	\$150k	30%

X Wrong way: simply average the rates across different sub-categories.

✓ Right way: add up numerators and denominators first, then divide.

Simpson's paradox: the overall rate can reverse every group's rate

Admission rate in two departments,
500 men and 500 women applying



Department	Men admitted / applied	Women admitted / applied
Engineering	10 / 100	80 / 400
History	240 / 400	80 / 100
Combined	250 / 500	160 / 500

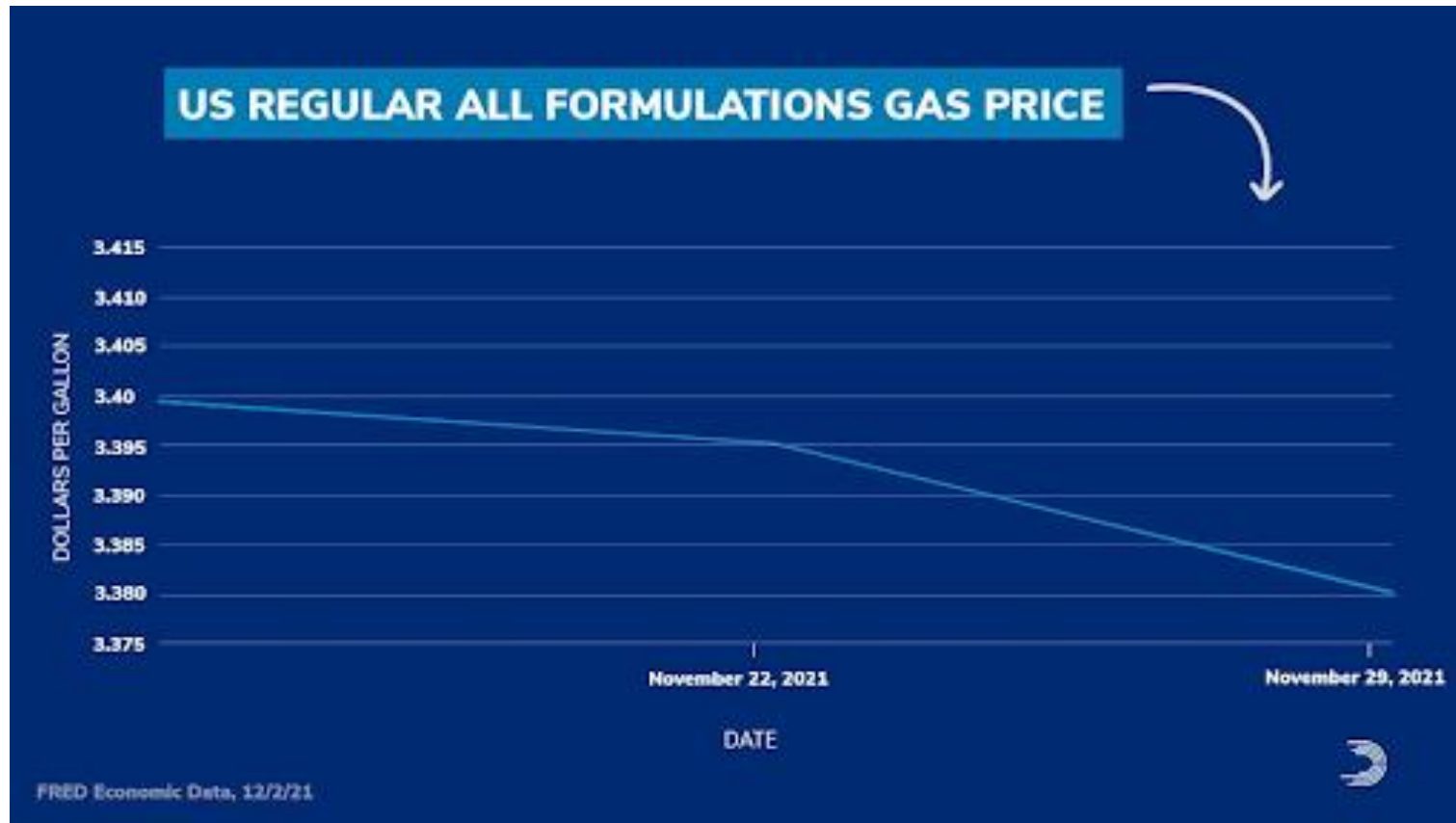
Rate reversal: Overall, the admission rate is higher for men, but women's rate is higher in each department. **Why?**

- Application gender mix differs across departments: More women applied to Engineering, which admits few.

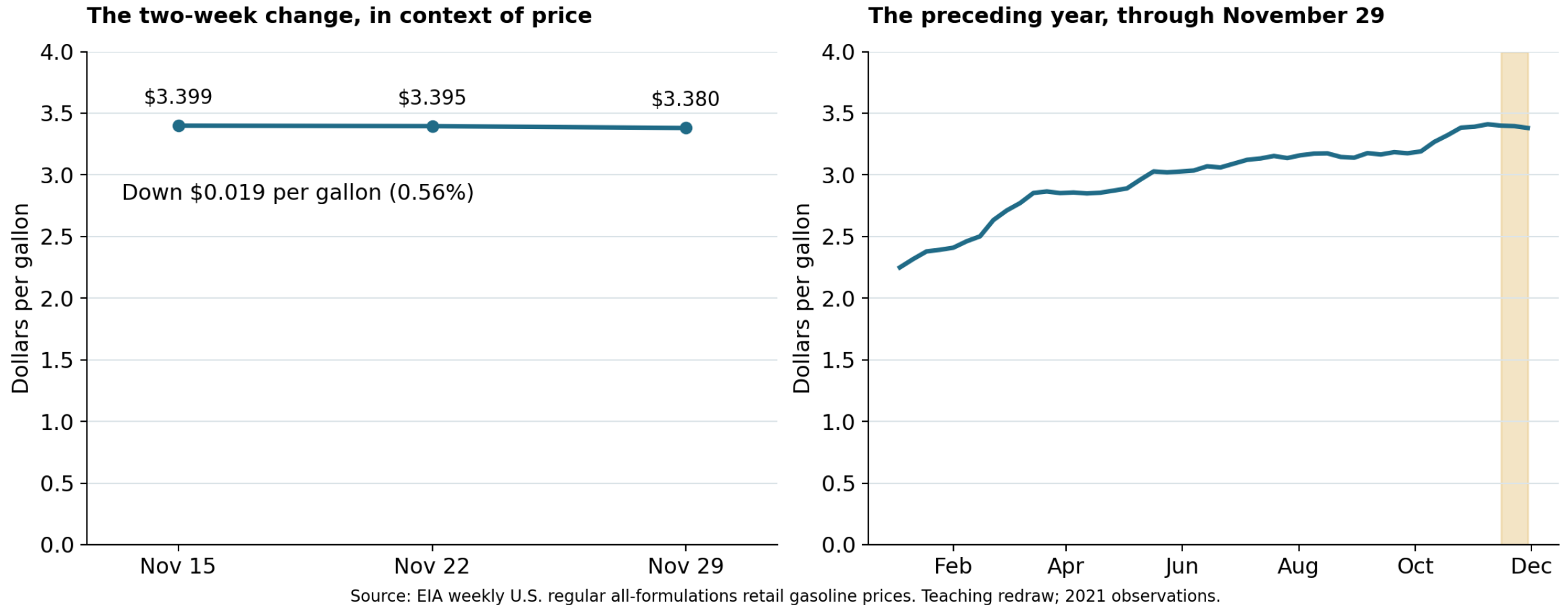
Be careful about what the "overall rate" tells you: when groups differ in mix, an overall rate can reverse what happens within each group. **Compare rates within groups first, and check each group's share before trusting the overall rate.**

Non-zero axis starts can be misleading

On December 2, 2021, the Democratic Congressional Campaign Committee (DCCC) tweeted this chart of U.S. gas prices with the message “Thanks, @JoeBiden.”



Same data, honest scale: a two-cent dip in a rising year



Summary

1. Pivot tables summarize raw rows by one, two or more grouping variables, with any statistic; pivot charts compare the results. Explore freely, then rank and highlight when you present.
2. An average hides the distribution. Describe the typical case (mean, median) and the tail (percentiles, tail share); see the shape with a histogram, box plot or percentile chart, and concentration with a Pareto chart or Lorenz curve.
3. Before concluding: look at outliers before excluding them (they move the mean, not the median); say which population you describe; add numerators and denominators before dividing, and compare rates within groups; check where the axis starts.